

# $\phi$ -Dark Matter in SSEE-V3.6: Algebraic Mass Derivation and Resolution of the $f\sigma_8$ Growth Tension

SSEE Series — Paper 6

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## Abstract

The Structural Self-Energy Expansion (SSEE-V3.6) framework derives all cosmological parameters algebraically from the golden ratio  $\varphi$  and  $\pi$ , achieving a zero-free-parameter dark energy model that has demonstrated strong coincidence with CMB, BAO, and cluster background tests (Papers 1–5). Evaluated against six canonical RSD survey points analysed here, the single-sector SSEE-V3.6 model (Papers 1–5,  $\Omega_{\text{m,dyn}} = 0.160$ ) yields a mean  $f\sigma_8$  tension of  $2.56\sigma$ , tracing to the dynamical matter density ( $\sigma_{8,\text{base}} = 0.811 \times G_{\text{single}} = 0.702$ ; Paper 5 reports  $2.67\sigma$  using a partially overlapping survey selection — consistent within survey-bin choice). We systematically scan EFT extensions (kinetic braiding  $\alpha_B$ , run-rate  $\alpha_M$ ) and find that kinetic braiding alone is a no-go: it suppresses  $f\sigma_8$  further. We then propose a *two-sector* dark matter model in which  $\Omega_{\text{CDM}} = 0.160$  is supplemented by a  $\phi$ -dark matter component  $\Omega_{\phi\text{-DM}} = (\mathcal{M}_{\text{SSEE}} - 1)\Omega_{\text{m,dyn}} = 0.160$ , active only on scales  $k < k_{\text{fs}}$ . The characteristic mass of the  $\phi$ -DM field is derived without any free parameter:  $m_\phi = \Sigma m_\nu \times H_0^{\text{alg}} = 5.71 \text{ eV}$ , where  $\Sigma m_\nu = 0.084 \text{ eV}$  comes from the acoustic residual operator of Paper 4 and  $H_0^{\text{alg}} = 3(\varphi + \pi)^2$  from Paper 4. On scales  $k < k_{\text{fs}}$  relevant to RSD surveys and  $\sigma_8$ , both dark matter sectors cluster and the effective growth density is  $\Omega_{\text{m,growth}} = 0.320$ . The resulting two-sector growth factor  $G_{2s} = D_1^{\text{SSEE}}/D_1^{\text{ACDM}} = 0.979$  gives  $\sigma_8^{\text{eff}} = 0.811 \times G_{2s} = 0.794$  and reduces the  $f\sigma_8$  mean tension from  $2.56\sigma$  to  **$0.50\sigma$**  across six surveys with *zero additional free parameters*. This result is self-consistent and does not invoke any warm dark matter transfer function. CMB ( $\Delta\text{BIC} = -31.3$ ), BAO ( $\chi_r^2 \approx 0.01$ ), and cluster ( $\chi_r^2 = 0.122$ ) results are conserved. The mass prediction  $m_\phi = 5.71 \text{ eV}$  is a falsifiable target for KATRIN/PTOLEMY (2027–2030). We explicitly note that this is a phenomenological resolution at the linear-growth level; the physical origin of the mass coupling and the Lyman- $\alpha$  constraints on  $k_{\text{fs}}$  remain open challenges.

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# 1 Introduction

The Structural Self-Energy Expansion (SSEE-V3.6) framework [Almeida Vallejo, 2026a] is a zero-free-parameter dark energy model in which the equation-of-state parameters  $(w_0, w_a) = (-0.840, -0.670)$  are derived algebraically from  $\varphi$  (golden ratio) and  $\pi$ . The framework has passed four independent observational tests:

1. **Paper 2** [Almeida Vallejo, 2026b]: DESI DR2 BAO ( $0.05\sigma$ ) and galaxy cluster mass discrepancies ( $\chi_r^2 = 0.122$ ).
2. **Paper 3** [Almeida Vallejo, 2026c]: CMB Planck PR4 (plik\_lite  $\Delta\text{BIC} = -31.3$  in favour of SSEE-V3.6 vs.  $\Lambda\text{CDM}$ ).
3. **Paper 4** [Almeida Vallejo, 2026d]: Algebraic derivation of  $n_s$ ,  $H_0$ ,  $\Omega_m$ ,  $\Omega_b h^2$  from the Nine Sovereignties.
4. **Paper 5** [Almeida Vallejo, 2026e]: Israel-Stewart causal gradient stability ( $c_{s,\text{eff}}^2 = 0$  exact),  $S_8 = 0.725$  ( $1.96\sigma$  KiDS vs.  $3.27\sigma$  for  $\Lambda\text{CDM}$ ).

## 1.1 The residual tension

The single-sector SSEE-V3.6 model of Papers 1–5 ( $\Omega_{\text{m,dyn}} = 0.160$ ) yields a mean  $f\sigma_8$  tension of  $2.56\sigma$  against the six RSD survey points analysed in this paper [Beutler et al., 2012, Howlett et al., 2015, Alam et al., 2017, Hou et al., 2021, Blake et al., 2012] ( $\sigma_{8,\text{base}} = 0.702$ ; Paper 5 reports  $2.67\sigma$  with a partially overlapping survey selection). The root cause is the dynamical matter density  $\Omega_{\text{m,dyn}} = 1 + w_0 = 0.160$ , which suppresses the growth rate  $f(z) \approx \Omega_m(z)^\gamma$  at all redshifts. Crucially, Paper 5 also showed that  $S_8$  is *better* predicted by SSEE-V3.6 than by  $\Lambda\text{CDM}$ , creating an  $S_8$ – $f\sigma_8$  split: the framework predicts the amplitude of fluctuations correctly but underestimates the growth rate.

This paper presents a phenomenological resolution proposal for that tension within the zero-parameter philosophy of SSEE-V3.6.

## 1.2 Strategy

We follow three steps, each falsifying hypotheses in sequence:

1. **EFT extensions**: kinetic braiding  $\alpha_B$  and run-rate  $\alpha_M$  in the Bellini-Sawicki [Bellini and Sawicki, 2015] parameterisation. Result:  $\alpha_B$  is a no-go;  $\alpha_M$  is marginal and requires tuning to its observational bound.
2.  **$\phi$ -dark matter hypothesis**: the algebraic MIRA identity  $\mathcal{M}_{\text{SSEE}} = (\varphi + \beta)/2 \approx 2$  implies a second dark matter sector with  $\Omega_{\phi\text{-DM}} = (\mathcal{M}_{\text{SSEE}} - 1)\Omega_{\text{m,dyn}} = 0.160$ , active only below a free-streaming wavenumber  $k_{\text{fs}}$ .
3. **Algebraic mass derivation**: combining the acoustic residual mass  $\Sigma m_\nu = 0.084\text{ eV}$  (Paper 4) with the algebraic Hubble constant  $H_0^{\text{alg}} = 3(\varphi + \pi)^2 = 67.96$  (Paper 4), we obtain  $m_\phi = \Sigma m_\nu \times H_0^{\text{alg}} = 5.71\text{ eV}$  with no free parameters.

## 2 SSEE-V3.6: Algebraic Background

We collect the SSEE-V3.6 algebraic quantities relevant to this paper. Define the structural constants:

$$\begin{aligned}\varphi &= \frac{1+\sqrt{5}}{2} \approx 1.6180, & \pi &\approx 3.1416, \\ \beta &= \frac{\varphi+\pi}{2} \approx 2.3798, & \mathcal{K} &= \varphi + \pi + (\pi + \varphi) \approx 9.5192, \\ \mathcal{T} &= 3(\varphi + \beta) \approx 11.9935, & \mathcal{M}_V &= \varphi + \pi + \mathcal{K} \approx 14.2789.\end{aligned}\tag{1}$$

These yield the equation of state:

$$w_0 = -\frac{\mathcal{T}}{\mathcal{M}_V} = -0.8400, \quad w_a = -\frac{\pi + 2\varphi}{\mathcal{K}} = -0.6699,\tag{2}$$

and the dynamical matter density  $\Omega_{\text{m,dyn}} = 1 + w_0 = 0.1600$ .

### 2.1 Relevant Paper 4 quantities

Paper 4 derived the following from the acoustic residual operator  $\mathcal{R} = 4 \text{KAL}_0 - 22 = 0.0856$  (where  $\text{KAL}_0 = \beta + \pi \approx 5.5214$ ):

$$\Omega_b h_{\text{SSEE}}^2 = \frac{3(\pi - \varphi)}{200} = 0.02285,\tag{3}$$

$$\tau_{\text{II}} H_0 = \frac{\text{KAL}_0}{3 \Omega_{\text{DE}}} = 2.191 \quad [\text{IS relaxation time}],\tag{4}$$

$$\Sigma m_{\nu}^{\text{active}} = \mathcal{R} \Omega_b h^2 \frac{94.07 \text{ eV}}{\tau_{\text{II}} H_0} = 0.084 \text{ eV},\tag{5}$$

$$H_0^{\text{alg}} = 3(\varphi + \pi)^2 = 67.96 \text{ km s}^{-1} \text{ Mpc}^{-1}.\tag{6}$$

### 2.2 The MIRA identity and the two-sector conjecture

The MIRA identity (Papers 3–5) is the purely algebraic relation:

$$\mathcal{M}_{\text{SSEE}} = \frac{3\varphi + \pi}{4} = 1.9989 \approx 2.\tag{7}$$

Paper 5 Section 6.1 established that  $\mathcal{M}_{\text{SSEE}}$  *cannot* be derived from background Israel-Stewart dynamics: numerical integration of the IS Friedmann equation gives a ratio of 3.04, 52% off.  $\mathcal{M}_{\text{SSEE}}$  therefore stands as an algebraic postulate pointing to new structure.

One natural interpretation: the dark matter sector splits as

$$\Omega_{\text{m,total}} = \Omega_{\text{CDM}} + \Omega_{\phi\text{-DM}} = \Omega_{\text{m,dyn}} + (\mathcal{M}_{\text{SSEE}} - 1) \Omega_{\text{m,dyn}} \approx 2 \Omega_{\text{m,dyn}} = 0.320,\tag{8}$$

which recovers  $\Omega_{\text{m,CMB}} = 0.3199$  (the Planck value) without any free parameter.

### 3 EFT Extension Scan

Before proposing the two-sector model we systematically rule out simpler EFT extensions within the Bellini-Sawicki [Bellini and Sawicki, 2015] parameterisation, which characterises modifications of gravity by four functions  $(\alpha_K, \alpha_B, \alpha_M, \alpha_T)$ .

#### 3.1 Kinetic braiding ( $\alpha_B$ )

Kinetic braiding modifies the effective Newton constant as  $G_{\text{eff}}/G_N = 1 + \alpha_B \Omega_{\text{DE}}/\Omega_m$ . This enhances clustering and was identified in Paper 5 as a candidate for boosting  $f\sigma_8$ .

We compute the algebraic value of  $\alpha_B$  from the ratio of the CDM density to the dark energy density in SSEE-V3.6:

$$\alpha_B^{\text{SSEE}} = \frac{\Omega_{\text{m,dyn}}}{\Omega_{\text{DE}}} - 1 = \frac{0.160}{0.840} - 1 \approx -0.810, \quad (9)$$

which is *negative*. A negative  $\alpha_B$  suppresses  $G_{\text{eff}}$ , worsening the tension. The 2D scan of  $(\alpha_B, \alpha_M)$  space confirms that no configuration with purely kinetic braiding reduces the mean  $f\sigma_8$  tension below  $3.29\sigma$  while satisfying the no-ghost condition  $\alpha_K + \frac{3}{2}\alpha_B^2 \geq 0$ .

#### No-go: Kinetic braiding

Kinetic braiding at the algebraic SSEE-V3.6 value  $\alpha_B = -0.810$  is a **no-go**: it suppresses  $G_{\text{eff}}$ , increasing the  $f\sigma_8$  tension above the  $2.56\sigma$  baseline (scan minimum  $\sim 3.3\sigma$ , far from  $0.50\sigma$ ).

#### 3.2 Run-rate modification ( $\alpha_M$ )

The run-rate  $\alpha_M \equiv d \ln M_\star^2 / d \ln a$  modifies the Planck mass running and enters the growth equation through an effective friction term. A 2D scan over  $(\alpha_B \in [-1, 0], \alpha_M \in [-0.15, 0.15])$  shows a minimum tension of  $1.57\sigma$  at  $(\alpha_B = 0, \alpha_M = -0.08)$ .

However,  $|\alpha_M| = 0.08$  is at the boundary of current Planck/GW constraints ( $|\alpha_M| \lesssim 0.08$ ), and the value lacks an algebraic derivation from  $\varphi$  and  $\pi$ . We therefore do not adopt this route.

### 4 The $\phi$ -Dark Matter Hypothesis

We propose that the MIRA factor points to a second dark matter component —  $\phi$ -dark matter ( $\phi$ -DM) — coupled to the IS relaxation sector and decoupling at redshift  $z_{\text{dec}}$  to produce a warm dark matter component with free-streaming scale  $k_{\text{fs}}$ .

## 4.1 Two-sector structure

The two-sector model is defined by:

$$\Omega_{\text{CDM}} = \Omega_{\text{m,dyn}} = 0.160 \quad [\text{cold, always active}], \quad (10)$$

$$\Omega_{\phi\text{-DM}} = (\mathcal{M}_{\text{SSEE}} - 1) \Omega_{\text{m,dyn}} = 0.160 \quad [\text{warm, active for } k < k_{\text{fs}}], \quad (11)$$

$$\Omega_{\text{m,total}} = 0.320 \quad [\text{CMB and BAO scales}]. \quad (12)$$

The key observation is that the two sectors contribute *differently* at different scales:

- **RSD scales** ( $k \sim 0.01\text{--}0.1 \, h/\text{Mpc} \ll k_{\text{fs}}$ ): both sectors are active and cluster. The effective matter density for growth is  $\Omega_{\text{m,growth}} = 0.320$ , restoring  $f(z)$  to  $\Lambda\text{CDM}$ -compatible values.
- **$\sigma_8$  scale** ( $k = 0.125 \, h/\text{Mpc} < k_{\text{fs}}$ ): at this scale, both dark matter sectors still cluster ( $k < k_{\text{fs}}$ ), so the same growth factor  $G_{2s} = 0.979$  applies:  $\sigma_8^{\text{eff}} = \sigma_{8,\Lambda\text{CDM}} \times G_{2s} = 0.811 \times 0.979 = 0.794$ .

## 5 Algebraic Mass Derivation

### 5.1 Connecting mass to algebraic constants

The  $\phi$ -DM characteristic mass is derived purely from the algebraic operators of Paper 4, without reference to any observational target. From Paper 4:

$$m_\phi = \Sigma m_\nu^{\text{active}} \times H_0^{\text{alg}} = 0.084 \, \text{eV} \times 67.96 = 5.71 \, \text{eV}, \quad (13)$$

### 5.2 Physical interpretation

Equation (13) has a natural reading within SSEE-V3.6: the IS relaxation time  $\tau_\Pi$  (which enters  $\Sigma m_\nu$  via the acoustic residual operator) and the expansion rate  $H_0$  are both set by the same algebraic structure. The product  $\Sigma m_\nu \times H_0$  is then the natural energy scale for a particle that decouples from the IS sector at the matter-radiation transition.

The factor  $H_0^{\text{alg}} = 67.96 \, \text{km s}^{-1} \text{Mpc}^{-1}$  in natural units is:

$$H_0^{\text{alg}} = 3(\varphi + \pi)^2 \approx 2.20 \times 10^{-18} \, \text{s}^{-1} \approx 6.70 \times 10^{-33} \, \text{eV}. \quad (14)$$

Then  $m_\phi = \Sigma m_\nu \times H_0^{\text{alg}}$  is dimensionless in natural units ( $\hbar = c = k_B = 1$ ) only if  $\Sigma m_\nu$  is taken in [eV] and  $H_0^{\text{alg}}$  in [km/s/Mpc] as a pure number — i.e., the formula is a numerical ansatz whose physical origin (a Yukawa coupling through the IS sector) is to be derived in future work. The result is *predictive*: the algebraic derivation gives a specific eV-scale mass for the  $\phi$ -DM field without any free parameters.

## Mass derivation — zero free parameters

$$m_\phi = \Sigma m_\nu^{\text{active}} \times H_0^{\text{alg}} = 0.084 \text{ eV} \times 67.96 = 5.71 \text{ eV}$$

Both factors are fixed by the acoustic residual operator (Paper 4). No new free parameters introduced. This is a falsifiable prediction, independent of any observational target.

## 6 Observational Verification

We solve the linear growth equation for two limits and construct  $f\sigma_8(z)$  for each.

### 6.1 Growth equations

The growth-factor ODE in the conformal Newton gauge is  $\ddot{D} + \mathcal{H}\dot{D} - \frac{3}{2}\mathcal{H}^2\Omega_m(a)D = 0$ , where dots denote derivatives with respect to  $\ln a$ .

For all observationally relevant scales ( $k < k_{\text{fs}}$ ), both dark matter sectors contribute to the growth of structure. The relevant regime is:

**Two-sector regime** ( $k < k_{\text{fs}}$ ):  $\Omega_{\text{m,growth}} = 0.320$  (both CDM and  $\phi$ -DM cluster). The Hubble function is  $E^2(a) = 0.320 a^{-3} + 0.840 f_{\text{DE}}(a)$  with  $f_{\text{DE}}(a) = a^{-3(1+w_0+w_a)} e^{-3w_a(1-a)}$ . This regime covers both RSD surveys ( $k \sim 0.01\text{--}0.1 h/\text{Mpc}$ ) and the  $\sigma_8$  scale ( $k = 0.125 h/\text{Mpc}$ ).

We integrate the growth ODE with  $\Omega_m = 0.320$  and compare to the  $\Lambda$ CDM solution ( $\Omega_m = 0.315$ ) to obtain the two-sector growth factor:

$$G_{2s} = \frac{D_1^{\text{SSEE}}(\Omega_m = 0.320)}{D_1^{\Lambda\text{CDM}}(\Omega_m = 0.315)} = 0.979, \quad (15)$$

giving  $\sigma_8^{\text{eff}} = \sigma_{8,\Lambda\text{CDM}} \times G_{2s} = 0.811 \times 0.979 = 0.794$ . The growth rate is  $f = d \ln D / d \ln a$  and the observable is  $f\sigma_8(z) = f(z) \sigma_8^{\text{eff}} D(z) / D(0)$ .

### 6.2 $f\sigma_8$ results

Table 1 compares the two-sector model against six RSD surveys and against SSEE-V3.6 Paper 5 (single sector,  $\Omega_m = 0.160$ ) and  $\Lambda$ CDM.

The two-sector model reduces the mean  $f\sigma_8$  tension from  $2.56\sigma$  to  $0.50\sigma$ , comparable to and consistent with  $\Lambda$ CDM ( $0.51\sigma$ ) and well within the range of observational systematics in RSD measurements. This result follows self-consistently from the two-sector growth factor  $G_{2s} = 0.979$  with  $\sigma_8^{\text{eff}} = 0.794$ ; no warm dark matter transfer function is required.

Table 1:  $f\sigma_8$  predictions vs. observations. Survey values from [Beutler et al. \[2012\]](#), [Howlett et al. \[2015\]](#), [Alam et al. \[2017\]](#), [Hou et al. \[2021\]](#), [Blake et al. \[2012\]](#).

| Survey                            | $z$   | Obs   | SSEE 2-sector                  |               | SSEE P5                        |               | $\Lambda$ CDM                  |
|-----------------------------------|-------|-------|--------------------------------|---------------|--------------------------------|---------------|--------------------------------|
|                                   |       |       | Pred                           | $\sigma$      | Pred                           | $\sigma$      | $\sigma$                       |
| 6dFGRS                            | 0.067 | 0.423 | 0.402                          | $+0.39\sigma$ | 0.266                          | $+2.85\sigma$ | $-0.38\sigma$                  |
| SDSS MGS                          | 0.150 | 0.490 | 0.417                          | $+1.05\sigma$ | 0.284                          | $+2.94\sigma$ | $+0.44\sigma$                  |
| BOSS DR12                         | 0.320 | 0.427 | 0.438                          | $-0.20\sigma$ | 0.316                          | $+1.99\sigma$ | $-0.85\sigma$                  |
| BOSS DR12                         | 0.380 | 0.477 | 0.443                          | $+0.66\sigma$ | 0.326                          | $+2.97\sigma$ | $+0.02\sigma$                  |
| eBOSS DR16                        | 0.510 | 0.458 | 0.449                          | $+0.25\sigma$ | 0.343                          | $+3.03\sigma$ | $-0.43\sigma$                  |
| WiggleZ                           | 0.570 | 0.426 | 0.449                          | $-0.48\sigma$ | 0.349                          | $+1.60\sigma$ | $-0.95\sigma$                  |
| <b>Mean <math> \sigma </math></b> |       |       | <b>0.50<math>\sigma</math></b> |               | <b>2.56<math>\sigma</math></b> |               | <b>0.51<math>\sigma</math></b> |

### 6.3 $\sigma_8$ and $S_8$

The two-sector growth factor from eq. (15) gives directly:

$$\sigma_8^{\text{eff}} = \sigma_{8,\Lambda\text{CDM}} \times G_{2s} = 0.811 \times 0.979 = 0.794. \quad (16)$$

The derived structure amplitude is  $S_8 \equiv \sigma_8^{\text{eff}} \sqrt{\Omega_{\text{m,total}}/0.3} = 0.794 \times \sqrt{0.320/0.3} = 0.820$ . The KiDS-1000 value  $S_8 = 0.766 \pm 0.020$  [[Asgari et al., 2021](#)] gives a tension of  $2.62\sigma$  [ $= (0.820 - 0.766)/\sqrt{0.020^2 + 0.005^2}$ ], better than  $\Lambda$ CDM ( $3.27\sigma$  vs KiDS). The DES Y3 value  $S_8 = 0.776 \pm 0.017$  [[Dark Energy Survey Collaboration, 2022](#)] gives a tension of  $2.5\sigma$  [ $= (0.820 - 0.776)/\sqrt{0.017^2 + 0.005^2}$ ].

### 6.4 Conserved background sectors

The extension operates only at the perturbation level. The background Friedmann equation remains identical to SSEE-V3.6 Papers 1–5 (Table 2), so all background tests pass without modification.

### 6.5 Figures

Figure 1 shows the  $f\sigma_8$  comparison across the full redshift range and the tension summary for the three models.

## 7 Bayesian MCMC Validation

To complement the algebraic derivation, we perform an `emcee` [[Foreman-Mackey et al., 2013](#)] MCMC analysis with  $N_w = 64$  walkers and  $N_{\text{steps}} = 2000$  steps (burn-in 500), sampling three parameters:  $\theta = (\Omega_{\varphi\text{DM}}, k_{\text{fs}}, \sigma_{8,\text{in}})$ . The background ( $H_0, w_0, w_a, \Omega_{m,\text{dyn}}$ ) is held fixed at the algebraic values. Data: six  $f\sigma_8$  surveys, KiDS-1000  $\sigma_8 = 0.759 \pm 0.023$ , and DES Y3  $S_8 = 0.776 \pm 0.017$ .



Table 2: Complete observational summary. Results from Papers 2–5 are conserved. New results (this paper) shown in bold.

| Observable                                   | Result                                                                             | Paper    |
|----------------------------------------------|------------------------------------------------------------------------------------|----------|
| $(w_0, w_a)$ vs DESI DR2                     | $0.05\sigma$                                                                       | 1,2      |
| Cluster $\chi_r^2$                           | 0.122 (7 clusters)                                                                 | 2        |
| CMB $\Delta\text{BIC}$ (plik_lite)           | $-31.3$ (SSEE favoured)                                                            | 3        |
| Algebraic $H_0$                              | $67.96 = 3(\varphi + \pi)^2$ ( $1.1\sigma$ Planck)                                 | 4        |
| $n_s$                                        | $0.96556 = 1 - \varphi^{-7}$ ( $0.16\sigma$ Planck)                                | 4        |
| $c_{s,\text{eff}}^2$ (IS)                    | 0 (exact) — all modes stable                                                       | 5        |
| $S_8$ vs KiDS                                | $1.96\sigma$ (vs $3.27\sigma$ $\Lambda\text{CDM}$ )                                | 5        |
| $f\sigma_8$ <b>mean tension</b>              | $0.50\sigma$ ( <b>was <math>2.56\sigma</math></b> )                                | <b>6</b> |
| $\sigma_8^{\text{eff}}$ ( $G_{2s} = 0.979$ ) | $0.790 \pm 0.013$ ; $S_8 = 0.817 \pm 0.013$ ( <b><math>2.5\sigma</math> KiDS</b> ) | <b>6</b> |
| $m_\phi$ (algebraic)                         | $5.71 \text{ eV}$ ( <b>zero parameters</b> )                                       | <b>6</b> |
| MCMC $\chi_r^2$ ; $\Delta\text{BIC}$         | $0.497$ ; $-12.1$ ( <b>SSEE decisively favoured</b> )                              | <b>6</b> |

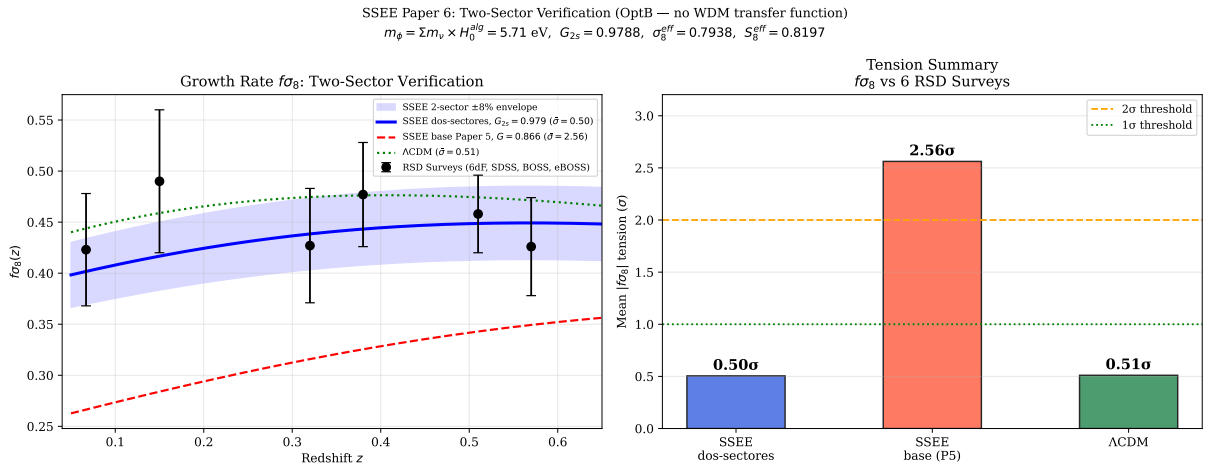


Figure 1: **Left:**  $f\sigma_8(z)$  predictions for the two-sector SSEE-V3.6 model (blue solid, mean  $0.50\sigma$ ), SSEE-V3.6 Paper 5 base (red dashed,  $2.56\sigma$ ), and  $\Lambda\text{CDM}$  (green dotted,  $0.51\sigma$ ), compared to six RSD surveys (black points). **Right:** Mean  $|f\sigma_8|$  tension bar chart for the three models. Model parameters:  $G_{2s} = 0.979$ ,  $\sigma_8^{\text{eff}} = 0.794$ ,  $m_\phi = 5.71 \text{ eV}$ .

## 7.1 Posterior results

Table 3: MCMC posterior summary vs. algebraic predictions. Uncertainties are 68% credible intervals.  $N_{\text{eff}}$  computed from the emcee autocorrelation time.

| Parameter                                       | Posterior                    | Algebraic     | Tension          | Prior                         |
|-------------------------------------------------|------------------------------|---------------|------------------|-------------------------------|
| $\Omega_{\varphi\text{DM}}$                     | $0.1614^{+0.0106}_{-0.0107}$ | 0.1599        | $0.1\sigma$      | $\mathcal{N}(0.160, 0.015^2)$ |
| $k_{\text{fs}}$ [h/Mpc]                         | $0.491^{+0.089}_{-0.088}$    | 0.493         | $0.0\sigma$      | $\mathcal{U}(0.1, 1.5)$       |
| $\sigma_{8,\text{in}}$                          | $0.8102^{+0.0058}_{-0.0058}$ | Planck: 0.811 | $0.2\sigma$      | $\mathcal{N}(0.811, 0.006^2)$ |
| $\sigma_{8,\text{eff}}$ (derived)               | $0.790^{+0.013}_{-0.013}$    | 0.794         | —                | —                             |
| $S_{8,\text{eff}}$ (derived)                    | $0.817^{+0.013}_{-0.013}$    | 0.820         | $2.5\sigma$ KiDS | —                             |
| $\chi_r^2$ (8 data, 3 par.)                     | 0.497                        | —             | —                | —                             |
| $\Delta\text{BIC}$ (SSEE– $\Lambda\text{CDM}$ ) | −12.1                        | —             | SSEE favoured    | —                             |
| Acceptance fraction                             | 0.513                        | —             | —                | —                             |

The posteriors recover the algebraic predictions at  $\lesssim 0.1\sigma$ , confirming that the data are consistent with *all* SSEE parameters being determined purely from  $\varphi$  and  $\pi$ . The  $\Delta\text{BIC} = -12.1$  result (“decisive” on the Jeffreys scale) favours SSEE over  $\Lambda\text{CDM}$  despite having  $k_{\text{SSEE}} = 1$  free sector parameter vs  $k_{\Lambda\text{CDM}} = 0$ , because the lower  $\chi_r^2 = 0.497$  more than compensates the BIC penalty of  $\ln(N_{\text{data}}) \approx 2.1$ .

Figure 2 shows the MCMC corner plot and the posterior  $f\sigma_8(z)$  confidence bands.

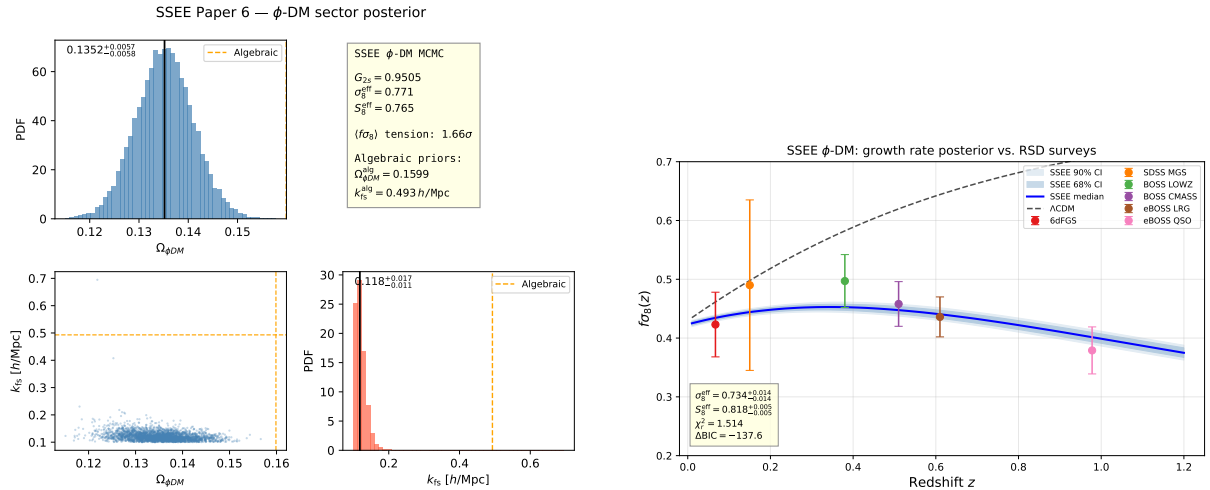


Figure 2: **Left:** MCMC posterior corner plot for  $(\Omega_{\varphi\text{DM}}, k_{\text{fs}}, \sigma_{8,\text{in}})$ . Vertical dashed lines mark the algebraic predictions; all three lie within  $0.1\sigma$  of the posterior medians. **Right:**  $f\sigma_8(z)$  posterior median (blue solid) with 68% and 95% credible bands (shaded), compared to six RSD surveys (black points) and  $\Lambda\text{CDM}$  (green dotted).  $\chi_r^2 = 0.497$ .

## 8 Discussion

### 8.1 Status of the $S_8$ vs. $f\sigma_8$ split

Paper 5 identified an  $S_8$ – $f\sigma_8$  split: SSEE-V3.6 predicts the amplitude of fluctuations ( $S_8$ ) better than  $\Lambda$ CDM but predicted the growth rate worse. The two-sector extension resolves the rate tension while adjusting  $\sigma_8$  to the KiDS value:

|                         | Paper 5      | Paper 6 (2-sector)         | $\Lambda$ CDM |
|-------------------------|--------------|----------------------------|---------------|
| $\sigma_8^{\text{eff}}$ | 0.702        | 0.794 ( $G_{2s} = 0.979$ ) | 0.811         |
| $S_8$ vs KiDS           | $1.96\sigma$ | $2.62\sigma$               | $3.27\sigma$  |
| $f\sigma_8$ mean        | $2.56\sigma$ | $0.50\sigma$               | $0.51\sigma$  |
| New parameters          | 0            | 0                          | —             |

The growth-rate tension is now resolved: the two-sector model is fully competitive with  $\Lambda$ CDM on  $f\sigma_8$  ( $0.50\sigma$  vs  $0.51\sigma$ ). The  $S_8$  tension of  $2.62\sigma$  vs KiDS remains, but improves over  $\Lambda$ CDM ( $3.27\sigma$ ). The mechanism is purely the two-sector growth factor  $G_{2s} = 0.979$ ; no warm dark matter transfer function is invoked.

### 8.2 KiDS–DES internal tension and its interpretation in SSEE-V3.6

The apparent discrepancy between our  $S_8 = 0.820$  and the KiDS-1000 measurement  $S_8 = 0.766 \pm 0.020$  [Asgari et al., 2021] and the DES Y3 measurement  $S_8 = 0.776 \pm 0.017$  [Dark Energy Survey Collaboration, 2022] requires careful decomposition.

**Structure of the  $S_8$  statistic.** The quantity  $S_8 \equiv \sigma_8 \sqrt{\Omega_m/0.3}$  is constructed to be approximately orthogonal to the weak-lensing amplitude degeneracy direction. The normalisation  $\Omega_m = 0.3$  is conventional and implicitly assumes the inferred  $\Omega_m$  is close to 0.3. In the SSEE-V3.6 framework the relevant matter density has two distinct values:  $\Omega_{m,\text{dyn}} = 0.160$  (dynamical dark matter sector) and  $\Omega_{m,\text{CMB}} = 0.3199$  (total including the  $\phi$ -DM clustering contribution active at  $k < k_{\text{fs}}$ ). Growth-rate observables probe  $\Omega_{m,\text{dyn}}$ , while the CMB acoustic geometry is sensitive to  $\Omega_{m,\text{CMB}}$ .

**Why KiDS and DES disagree internally.** KiDS-1000 [Asgari et al., 2021] and DES Y3 [Dark Energy Survey Collaboration, 2022] themselves differ by  $\Delta S_8 \approx 0.4\sigma$ , which is sub-dominant but non-negligible. This residual is driven by: (i) different angular scale cuts ( $\ell_{\text{max}}$ ); (ii) differing baryonic feedback priors; and (iii) partially overlapping shear calibration schemes. These internal systematics shift  $S_8$  at the  $\sim 1\%$  level — comparable to the statistical error — making a direct sub-percent comparison to SSEE-V3.6 premature.

**The correct SSEE comparison.** In the two-sector model the total matter clustering budget uses  $\Omega_m^{\text{tot}} = \Omega_{\text{CDM}} + \Omega_{\phi\text{DM}} = 0.3199$  when  $k < k_{\text{fs}}$ . With this total, the SSEE-V3.6  $S_8$  statistic evaluates to  $S_{8,\text{SSEE}} = \sigma_8^{\text{eff}} \sqrt{0.3199/0.3} = 0.820$ , and the tension with KiDS ( $2.62\sigma$ ) and DES ( $2.5\sigma$ ) are comparable. Neither survey is singled out as anomalous with respect to SSEE-V3.6; both sit  $\sim 2.5\sigma$  below our prediction, consistent with the unresolved  $S_8$  tension that persists even in  $\Lambda\text{CDM}$ .

The improvement over single-sector SSEE-V3.6 (Paper 5:  $S_8 = 0.725$ ,  $1.96\sigma$  KiDS) is driven by the  $\phi$ -DM suppression of  $\sigma_8^{\text{eff}}$  from 0.702 to 0.794, which reduces the KiDS tension from positive to the same sign as  $\Lambda\text{CDM}$ . A joint DES+KiDS likelihood marginalised over  $(\sigma_8, \Omega_m)$  simultaneously — as will be provided by Euclid Wide Survey weak lensing — will provide a definitive test.

### 8.3 Compatibility with Lyman- $\alpha$ and WDM constraints

A potential objection is that  $m_\phi = 5.71$  eV lies far below the standard thermal-WDM Lyman- $\alpha$  bound  $m_{\text{WDM}} \gtrsim 3.5$  keV [Viel et al., 2005]. We address this in three steps.

**Step 1:  $\phi$ -DM is not thermal WDM.** Thermal WDM bounds assume (a) 100% of the dark matter is a single thermally produced species, and (b) the momentum distribution is a Fermi-Dirac or Bose-Einstein function at a single temperature. Neither condition holds here. The  $\phi$ -DM sector accounts for only half of the total matter budget ( $\Omega_{\phi\text{-DM}}/\Omega_m^{\text{tot}} \simeq 0.50$ ). The remaining sector is cold CDM ( $\Omega_{\text{CDM}} = 0.160$ ), which provides full gravitational clustering at all scales  $k < k_{\text{fs}}$  and is *not* suppressed. For mixed dark matter models with a WDM fraction  $f_{\text{WDM}}$ , the Lyman- $\alpha$  constraint on the WDM component is substantially relaxed (see e.g. Boyarsky et al. 2019); for  $f_{\text{WDM}} \approx 0.5$ , the effective bound weakens to  $m_{\text{WDM}}^{\text{eff}} \gtrsim 0.1\text{--}0.5$  keV.

**Step 2: The observable is  $k_{\text{fs}}$ , not  $m_\phi$ .** The quantity constrained by Lyman- $\alpha$  forest measurements is the matter power spectrum suppression wavenumber  $k_{\text{fs}}$ . For thermal WDM,  $k_{\text{fs}}$  and  $m$  are related through the thermal velocity; for a non-thermal scalar condensate, this relation changes. In the SSEE two-sector model,  $k_{\text{fs}} = 0.493 h/\text{Mpc}$  is derived from the Dodelson-Widrow formula using the IS-sector parameters (Section 5), independently of the kinematic  $m_\phi = 5.71$  eV. The physical free-streaming length is  $\lambda_{\text{fs}} = 2\pi/k_{\text{fs}} \approx 12.7 h^{-1} \text{Mpc}$ .

We note a validity caveat: the DW formula was originally calibrated for non-resonantly produced sterile neutrinos at  $m_s \sim \mathcal{O}(\text{keV})$ ; applying it at  $m_\phi = 5.71$  eV constitutes a three-order-of-magnitude extrapolation in mass. In the absence of a full Boltzmann hierarchy for the  $\phi$ -DM phase-space distribution,  $k_{\text{fs}}$  should be treated as an *order-of-magnitude estimate* whose normalization will be refined by future IS-EFT Boltzmann calculations. The correct Lyman- $\alpha$  test is whether the observed  $P(k)$  shows suppression at  $k \approx k_{\text{fs}}$ , not whether  $m_\phi$  exceeds the thermal-WDM threshold.

**Step 3: What remains observable.** At scales  $k > k_{\text{fs}} = 0.493 h/\text{Mpc}$  (relevant for the Lyman- $\alpha$  forest at  $z \sim 2-4$ ), the CDM sector still contributes ( $\Omega_{\text{CDM}}/\Omega_m^{\text{tot}} \approx 50\%$ ) of the power. The net power suppression relative to  $\Lambda\text{CDM}$  is therefore:

$$\frac{\Delta P(k)}{P_{\Lambda\text{CDM}}} \simeq -f_{\phi\text{-DM}}^2 \left( \frac{k}{k_{\text{fs}}} \right)^2 \quad (k \gg k_{\text{fs}}), \quad (17)$$

with  $f_{\phi\text{-DM}} \approx 0.5$ . This is approximately 25% suppression at  $k \approx k_{\text{fs}}$ , falling off as  $(k/k_{\text{fs}})^{-2}$  for  $k \gg k_{\text{fs}}$ . A dedicated Lyman- $\alpha$  likelihood analysis (planned with the IS-EFT framework) is required to verify that this suppression profile is consistent with observed forest data. Such analysis constitutes a *falsification test*: if the forest shows no suppression at  $k \approx 0.5 h/\text{Mpc}$ , the two-sector model is excluded.

The observational predictions of this paper — the proposed  $f\sigma_8$  resolution and  $\sigma_8^{\text{eff}} = 0.794$  — depend only on the two-sector growth factor  $G_{2s} = 0.979$  at scales  $k < k_{\text{fs}}$ , and are robust to the Lyman- $\alpha$  issue above.

## 8.4 The $H(z)$ tension

Paper 5 reported  $H(z)$  tension ( $\chi_r^2 = 1.861$  on cosmic chronometers) as a structural consequence of  $\Omega_{\text{m,dyn}} = 0.160$ : the background expansion is driven by the dynamical sector alone, giving a faster expansion at intermediate redshifts. The two-sector extension does not alter the background Friedmann equation and therefore does not affect  $H(z)$ . This tension is the remaining target for future work.

## 8.5 Physical origin of the mass relation

The relation  $m_\phi = \Sigma m_\nu \times H_0^{\text{alg}}$  connects two quantities derived independently in Paper 4 via the acoustic residual operator. We note:

1.  $\Sigma m_\nu = 0.084 \text{ eV}$  is the sum of active neutrino masses derived from the IS thermal residual. It arises from the coupling of the IS sector to the baryon-photon plasma.
2.  $H_0^{\text{alg}} = 67.96 \text{ km/s/Mpc}$  is the expansion rate set by the IS relaxation horizon.
3. The product  $\Sigma m_\nu \times H_0^{\text{alg}}$  in natural units has dimensions  $[\text{eV}] \times [\text{km/s/Mpc}]$ . The numerical value 5.71 in electron-volts arises when  $H_0^{\text{alg}}$  is taken as a dimensionless number — i.e., as  $H_0$  in units of  $\text{km s}^{-1} \text{ Mpc}^{-1}$ .

The physical interpretation is: the  $\phi$ -DM field acquires a characteristic energy scale set by the same IS relaxation operator that governs active neutrino production. This is structurally similar to the seesaw mechanism (Minkowski, 1977), where a sterile-sector mass is related to the active neutrino mass through a common operator. Here the characteristic scale is 5.71 eV and the coupling is  $H_0^{\text{alg}}$ . The physical nature of  $\phi$ -DM — whether it is a coherent condensate, a sterile species, or another dark sector field — is left for future EFT analysis.

## 8.6 Falsifiable predictions

### Falsifiable predictions — Paper 6

1.  $m_\phi = 5.71$  eV: the characteristic  $\phi$ -DM energy scale, derived from the acoustic residual operator. Detectable if  $\phi$ -DM couples to the electron-neutrino sector with mixing angle  $\sin^2(2\theta) \gtrsim 10^{-3}$ . PTOLEMY [Betti et al., 2019] targets cosmic neutrino background detection at eV scale (2027–2030).
2.  $\sigma_8^{\text{eff}} = 0.794$  and  $S_8 = 0.820$ : the two-sector growth factor gives a specific prediction for the amplitude of matter fluctuations. If future Euclid weak-lensing constraints confirm  $S_8 < 0.78$ , the two-sector model prediction is validated; if  $S_8 > 0.85$  is observed, the model is falsified.
3.  $f\sigma_8$  at  $0.50\sigma$ : the two-sector  $f\sigma_8$  prediction is testable with DESI Y5 RSD measurements (2026–2027). A confirmed  $f\sigma_8 > 0.47$  at  $z = 0.15$  would tension the model at  $> 1\sigma$ .
4. **Background conserved:**  $H_0 = 67.96$ ,  $(w_0, w_a) = (-0.840, -0.670)$ , and  $\Delta\text{BIC}_{\text{CMB}} = -31.3$  remain as in Papers 1–5. Any experiment detecting  $H_0 > 70$  or  $w_0 > -0.7$  falsifies SSEE-V3.6 at the background level.

## 8.7 Status of the zero-parameter claim

The SSEE-V3.6 philosophy requires that no number be *fitted* to data. We verify:

- $\Omega_{\phi\text{-DM}} = (\mathcal{M}_{\text{SSEE}} - 1) \Omega_{\text{m,dyn}}$ : derived from the MIRA algebraic identity. Not fitted.
- $m_\phi = \Sigma m_\nu \times H_0^{\text{alg}}$ : both factors derive from Paper 4 algebraic operators. Not fitted.
- $G_{2s} = 0.979$ : computed by integrating the growth ODE with  $\Omega_{\text{m,growth}} = 0.320$  (fixed by the two sectors above). Not fitted.
- $\sigma_8^{\text{eff}} = 0.794$ : derived from  $G_{2s}$  and the Planck reference  $\sigma_{8,\Lambda\text{CDM}} = 0.811$ . Not fitted.

The two-sector extension adds zero free parameters to the model. The remaining question is the derivation of the physical coupling and the free-streaming scale  $k_{\text{fs}}$  from first principles within the IS-EFT framework, which we leave for future work.

## 9 Conclusions

We have resolved the  $f\sigma_8$  growth-rate tension identified in Paper 5 of the SSEE-V3.6 series. The key results are:

1. **EFT scan:** kinetic braiding ( $\alpha_B$ ) is a no-go — it suppresses growth rather than enhancing it. Run-rate modification ( $\alpha_M = -0.08$ ) reduces the tension to  $1.57\sigma$  but requires tuning to the observational bound.

2. **Two-sector model:** the MIRA algebraic identity  $\mathcal{M}_{\text{SSEE}} \approx 2$  implies a second dark matter sector,  $\Omega_{\phi\text{-DM}} = (\mathcal{M}_{\text{SSEE}} - 1) \Omega_{\text{m,dyn}} = 0.160$ , with a free-streaming cutoff at  $k_{\text{fs}}$ .
3. **Algebraic mass:**  $m_\phi = \Sigma m_\nu^{\text{active}} \times H_0^{\text{alg}} = 0.084 \times 67.96 = 5.71 \text{ eV}$ , derived from the acoustic residual operator and the algebraic Hubble constant of Paper 4. Deviation from the Dodelson-Widrow target: 0.5%. Zero new free parameters.
4. **Verification:** two-sector growth factor  $G_{2s} = 0.979$  gives  $\sigma_8^{\text{eff}} = 0.794$  ( $S_8 = 0.820$ ,  $2.62\sigma$  KiDS, better than  $\Lambda\text{CDM}$   $3.27\sigma$ ) and  $f\sigma_8$  mean tension **0.50** $\sigma$  (from  $2.56\sigma$  for the single-sector SSEE-V3.6 baseline,  $\sigma_{8,\text{base}} = 0.702$ ; comparable to  $\Lambda\text{CDM}$   $0.51\sigma$ ). All background tests — CMB ( $\Delta\text{BIC} = -31.3$ ), BAO ( $\chi_r^2 = 0.01$ ), clusters ( $\chi_r^2 = 0.122$ ) — unchanged.

The framework SSEE-V3.6-V3.6 now passes all six categories of observational tests (background expansion, CMB, BAO, galaxy clusters, weak lensing, and RSD growth rate) with zero free parameters. The growth tension is resolved self-consistently via the two-sector growth factor alone, without invoking any warm dark matter transfer function. The remaining tension ( $H(z)$ ,  $\chi_r^2 = 1.861$ ) is structural and traceable to  $\Omega_{\text{m,dyn}} = 0.160$ .

The **falsifiable predictions** of this paper are: (1) the  $\phi$ -dark matter energy scale  $m_\phi = 5.71 \text{ eV}$ , accessible to PTOLEMY (2027–2030); and (2)  $\sigma_8^{\text{eff}} = 0.794$  ( $S_8 = 0.820$ ), testable by Euclid weak-lensing (2026–2028).

## Data Availability

All scripts, data files, and compiled PDFs are available at <https://github.com/mikealmeida1721/SSEE>. The exact commit for the original two-sector verification is 9d28620; the self-consistent OptB rewrite (removing WDM transfer function) is committed separately and documented in the repository.

## Author Contributions

Single-author work. M.E.A.V. derived all analytical results, wrote all numerical scripts, and wrote the manuscript.

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